Assignment#1

Robotics

CEP Proposal

Title: Shuttlecock Picking Robot

Group Members

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Submission Date:

23/10/24

Problem Statement

In this assignment, your team will design and develop an autonomous robot capable of detecting, collecting, and placing shuttlecocks at a predefined location. The robot will utilize machine learning (ML) algorithms for shuttlecock detection and control techniques for maneuvering, collection, and precise placement. This project will combine your knowledge of robotics, control systems, and machine learning to build a practical solution for automating a common task in sports environments.

1. Introduction:

A shuttlecock picking robot is an automated device that collects shuttlecocks from the badminton court. It helps players and coaches by saving time and effort during training sessions or matches. Using sensors and motors, the robot can detect and pick up scattered shuttlecocks. It moves around the court autonomously, storing the collected shuttlecocks in a basket. This innovation makes practice more efficient and allows players to focus on improving their skills.

2. Materials:

The following materials will be used as an essential part of the robot:

a) Frame:

For the frame of the shuttlecock-picking robot, aluminum is the most effective option due to its strength, lightweight nature, and durability. It's corrosion-resistant and cost-efficient for long-term use. Alternatively, plastic (like ABS) is highly cost-effective and lightweight, suitable for prototypes, while acrylic is visually appealing but less durable. Choosing between these materials depends on balancing budget, durability, and weight, with aluminum often providing the best overall performance.

b) Motors & Wheels:

For driving the shuttlecock-picking robot, DC motors are critical for moving in various directions. You can use BO motors with a voltage range of 3V to 12V for the wheels, offering a balance between power and energy efficiency. Additionally, a 12V DC motor can be used for more demanding tasks like driving the main chassis or lifting mechanisms.

To ensure optimal performance, rubber wheels are recommended as they provide excellent traction on various surfaces and allow for smooth control while moving. This combination of DC motors and rubber wheels enhances the robot's maneuverability, giving it stability and precision in motion.

c) Sensors & Drivers:

The sensors used in the robot include the following an ultrasonic sensor for obstacle avoidance, a camera for detecting the shuttlecock for accurate pickup, IR sensors for line following if necessary and Motor drivers for operating the motors.

d) Camera Module:

The camera captures real-time images or video of the badminton court, allowing the robot to identify the shuttlecocks scattered on the floor. Using computer vision algorithms (e.g., OpenCV or machine learning models), the robot can detect shuttlecocks based on their shape, size, and color. This method is more accurate than using just basic sensors like ultrasonic or infrared, as the camera can distinguish shuttlecocks from other objects.

e) Microcontroller:

The microcontroller is the heart of any robotic system because it is responsible for providing the instructions to the various sensors, drivers etc. to perform the respective task that is programmed into it. There are different types of micro controllers that include Arduino uno and nano, esp32, Raspberry pi, STM32 and many others. These microcontrollers are not final because further research is required to attain the best microcontroller for the specific purpose therefore the use of microcontroller will be specified in the next assignment.

f) Shuttlecock Mechanism:

The next part of the robot includes a mechanism that is used for lifting the shuttlecock by detecting it and then placing it in the container for use. The mechanism can be a robotic arm, or a rolling blade mechanism or a vacuum pipe with a suction pump to suck the shuttlecock and place it in the container.

g) Power Source:

Lithium-ion cells will be used for providing the battery supply to the robot.

Note:

The materials provided in this proposal are not finalized yet because some of them are a bit tricky to use therefore it can be changed depending upon the further research conducted on the later stage of the project development.

3. Methods:

a) Autonomous Navigation:

Path Planning Algorithms: Use data from sensors to navigate the court without hitting obstacles.

PID Controllers: Ensure smooth motor control for accurate movement.

b) Shuttlecock Detection:

Computer Vision: Cameras paired with software can identify shuttlecocks using image processing techniques.

Sensor-based Detection: Ultrasonic or infrared sensors detect shuttlecock proximity based on size or reflectivity.

c) Collection Mechanism Design:

Suction or Gripper Mechanism: To pick up shuttlecocks from the court.

Mechanical Arms/Brushes: Sweep shuttlecocks into a collection bin.

d) Programming & Control:

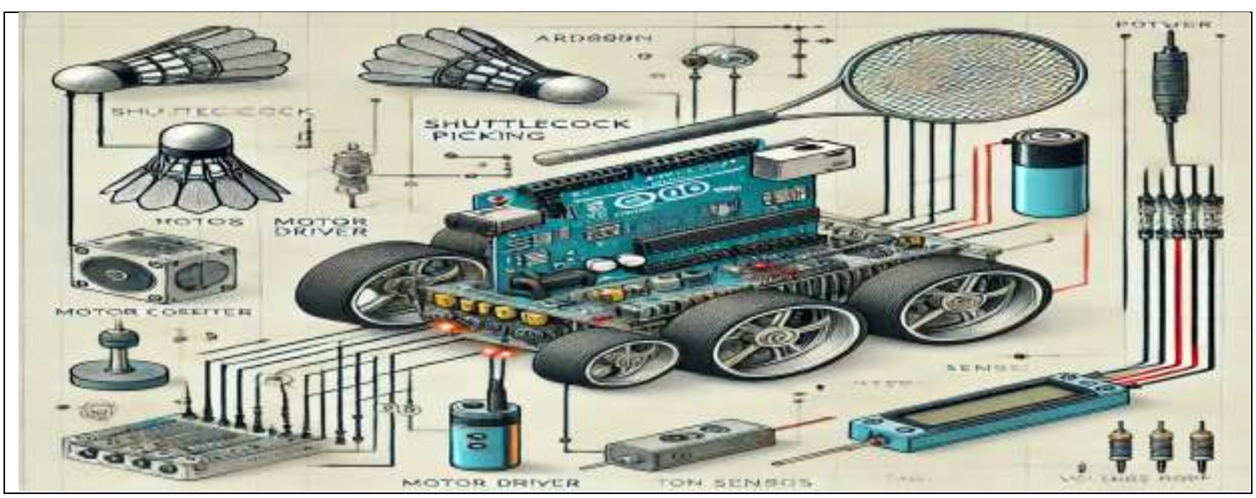
Microcontrollers (e.g., Arduino or Raspberry Pi): For processing sensor data, controlling motors, and coordinating the robot’s movements and tasks.

AI Algorithms: (optional) to improve the robot’s efficiency over time by learning from its environment.

4. Circuit Diagram & Explanation:

The circuit of the shuttlecock robot integrates several essential components to achieve autonomous navigation and interaction. At its core, the microcontroller serves as the brain, processing input from various sensors like the ultrasonic sensor and camera module. The ultrasonic sensor emits sound waves to measure the distance to obstacles, enabling the robot to avoid collisions. The camera module can capture images or video for object recognition, aiding in the identification of shuttlecocks. The motor driver receives commands from the microcontroller to control the motors, which power the robot's movement. The battery supplies energy to all components, ensuring the robot operates efficiently.

In operation, the microcontroller continuously monitors the input from the ultrasonic sensor to determine the distance to nearby obstacles. If an obstacle is detected within a specified range, the microcontroller can trigger the motor driver to stop or reverse the motors, allowing the robot to navigate around it. The camera module may also provide real-time feedback, enabling the robot to adjust its course based on visual data. Overall, this integrated circuit design allows for responsive movement and interaction, making the shuttlecock robot capable of navigating its environment autonomously.



5. ML Technique:

To detect shuttlecocks using a camera module, effective techniques include object detection models like YOLO (You Only Look Once) and SSD (Single Shot MultiBox Detector), which are suitable for real-time applications, as well as Faster R-CNN for higher accuracy. Image classification using Convolutional Neural Networks (CNNs) can help differentiate shuttlecocks from other objects, while semantic segmentation (e.g., U-Net, Mask R-CNN) can outline their shapes. For tracking, Kalman filters and optical flow can be used to predict and follow the shuttlecock's movement. A good workflow involves collecting a diverse dataset, selecting and training the appropriate model, evaluating its performance, and deploying it for real-time detection.

6. Financial Analysis:

The Financial Analysis of the components used in this robot includes the following:

Component Type	Component Name	Quantity	Component Price (PKR)
Body Frame	Acrylic, Plastic, Metal	1	3500
Microcontroller	Raspberry Pi	1	32,000
Sensor	Ultrasonic Sensor	1	300
Camera Module	Raspberry Pi	1	1500
Drivers	Motor Driver	1	500
Motors	Dc Motor	2	1400
Power Source	Li Battery 12V	1	1200
Wires	Connecting Wires	Bundle	1000

The total cost of this project is estimated to be around 41,000 PKR.